Why Are Customers Leaving?

A Data-Driven Analysis of Bank Customer Churn

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Dataset	Customers	Variables	Churn Rate	Missing Values
Churn_Modelling.csv	10,000	11 (after cleaning)	20.4%	None — 0

Introduction

Imagine walking into a bank branch and learning that 1 in 5 customers has already decided to close their account and leave. That is exactly what this dataset reveals. With 10,000 bank customers across France, Germany, and Spain, this analysis explores a question that every bank manager cares deeply about: who is leaving, and why?

Customer churn — the term used when a customer ends their relationship with a bank — is one of the most expensive problems in the financial industry. It costs roughly five times more to acquire a new customer than to retain an existing one. Understanding which customers are at risk allows banks to act early, offer targeted incentives, and significantly reduce revenue losses.

This analysis was conducted entirely in R using the tidyverse framework. Eight visualizations and a comparative risk profile were produced to answer four core data questions that guide this investigation.

Data Questions

Before beginning the analysis, four questions were developed to give direction to the exploration:

#	Question
Q1	What is the overall churn rate and how serious is the problem?
Q2	Which countries and demographics are most at risk of churning?
Q3	Does account activity and product usage affect whether a customer leaves?
Q4	What does the profile of a high-risk customer look like?

Dataset Overview

The dataset contains records for 10,000 bank customers. After cleaning, 11 variables were retained for analysis. Three columns were removed — row number, customer ID, and surname — because they are simply identification labels that tell us nothing meaningful about customer behavior. The key variable in this dataset is 'Exited', which tells us whether each customer closed their account (Churned) or stayed.

Table 1. Key Variables Used in Analysis

Variable	Type	Description
credit_score	Numeric	Customer's credit rating (350 to 850)

geography	Category	Country — France, Germany, or Spain
gender	Category	Male or Female
age	Numeric	Customer age in years (18 to 92)
tenure	Numeric	Years as a bank customer (0 to 10)
balance	Numeric	Current account balance in USD
num_of_products	Category	Number of bank products held (1 to 4)
has_cr_card	Category	Whether customer holds a credit card
is_active_member	Category	Whether customer is actively using services
estimated_salary	Numeric	Annual salary estimate in USD
exited	Category	Target: Churned or Stayed

Finding 1 — The Scale of the Problem

The first chart below answers our most fundamental question: how many customers are actually leaving? The answer is more serious than many banks would be comfortable admitting.

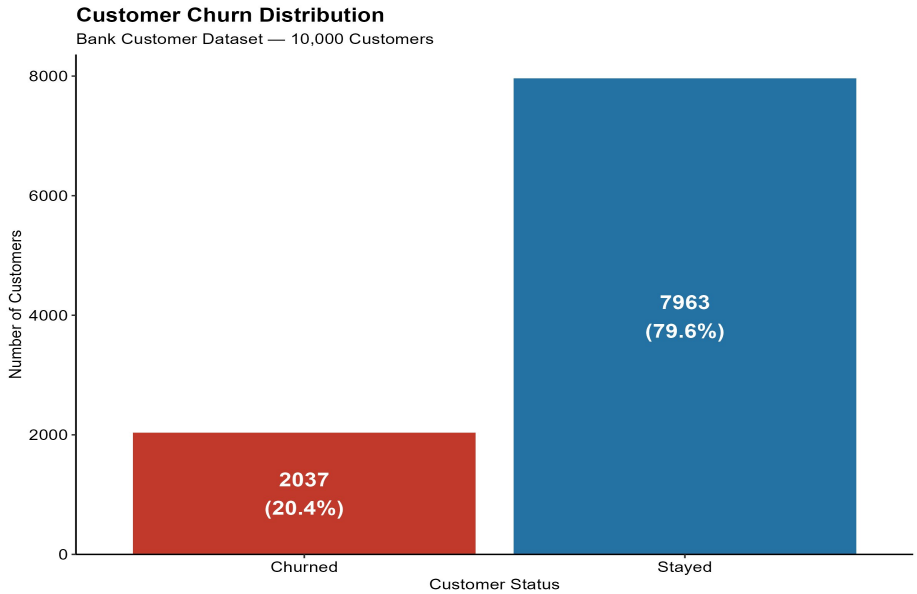

Figure 1. Overall customer churn distribution across 10,000 bank customers.

Out of 10,000 customers, 2,037 — or exactly 20.4% — have already left the bank. That means 1 in every 5 customers walks out the door. While 79.6% remain loyal, a 20% churn rate is widely considered a serious warning sign in banking. Industry benchmarks suggest healthy churn should sit below 5-10%. This bank is operating at double or quadruple that threshold.

> *To put this in financial terms: if each customer generates an average of $1,000 in annual revenue, 2,037 churned customers represent roughly $2 million in lost annual income — before accounting for the cost of replacement.*

Finding 2 — Germany Has a Serious Problem

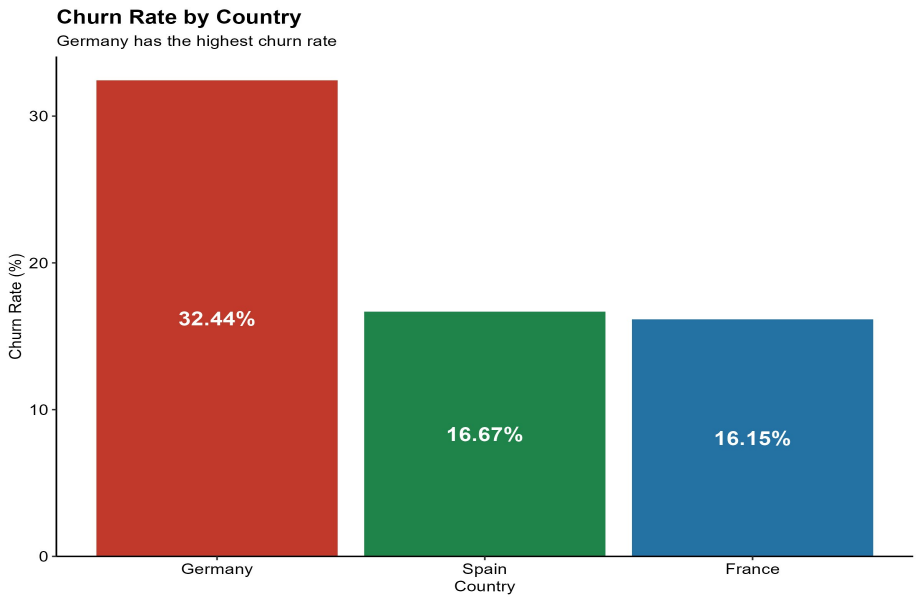

Figure 2. Churn rate by country — only churned customers shown.

When we break the churn rate down by country, Germany stands out dramatically. German customers churn at 32.44% — nearly double the rate of France (16.15%) and Spain (16.67%). This is not a small difference. It suggests something fundamentally different is happening in the German market: perhaps greater competition from local banks, differences in customer service quality, or cultural differences in banking loyalty.

> *This finding should prompt immediate investigation by the bank's regional management team. Germany accounts for 25% of the total customer base (2,509 customers) but is responsible for a disproportionate share of the churn problem.*

Finding 3 — Age is the Strongest Predictor

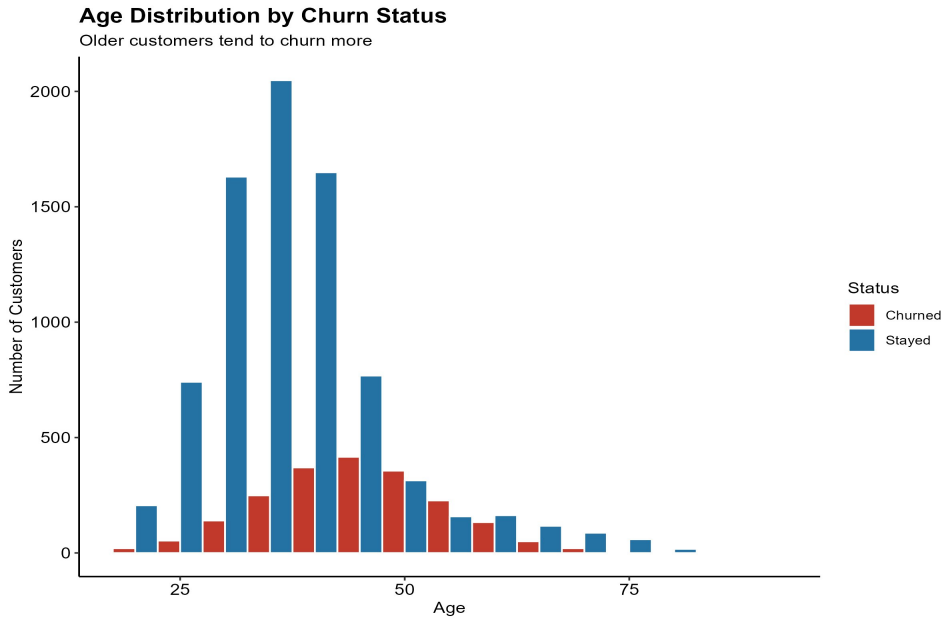

Figure 3. Age distribution comparing customers who churned versus stayed.

The age chart tells a very clear story. Customers who stay tend to be younger — mostly in their late 20s to early 40s. Customers who churn are noticeably older, with the churned red bars becoming relatively larger compared to the blue stayed bars as we move past age 45. The average age of a churned customer is 44.8 years, compared to just 37.4 years for customers who stay.

> *This 7-year age gap is statistically meaningful. Middle-aged and older customers may be consolidating their finances, retiring, or moving to wealth management services elsewhere. Banks should consider dedicated retention programs for customers entering the 40 to 60 age bracket.*

Finding 4 — Churned Customers Have More Money

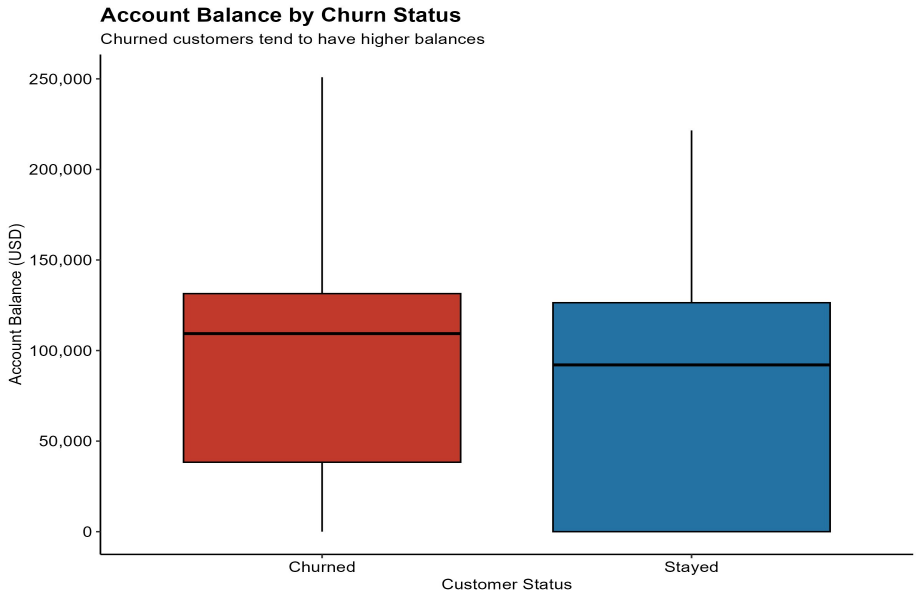

Figure 4. Account balance comparison between churned and stayed customers.

This finding is counterintuitive and important. Churned customers actually have higher average account balances ($91,109) compared to customers who stayed ($72,745). The boxplot confirms this — the churned customers box sits noticeably higher on the y-axis. This means the bank is not just losing customers — it is losing its most financially valuable customers.

> *The implication is significant: retention efforts should not focus only on low-value customers. High-balance customers are quietly leaving, potentially taking their savings to competitor banks or investment platforms.*

Finding 5 — Inactive Members Leave More Often

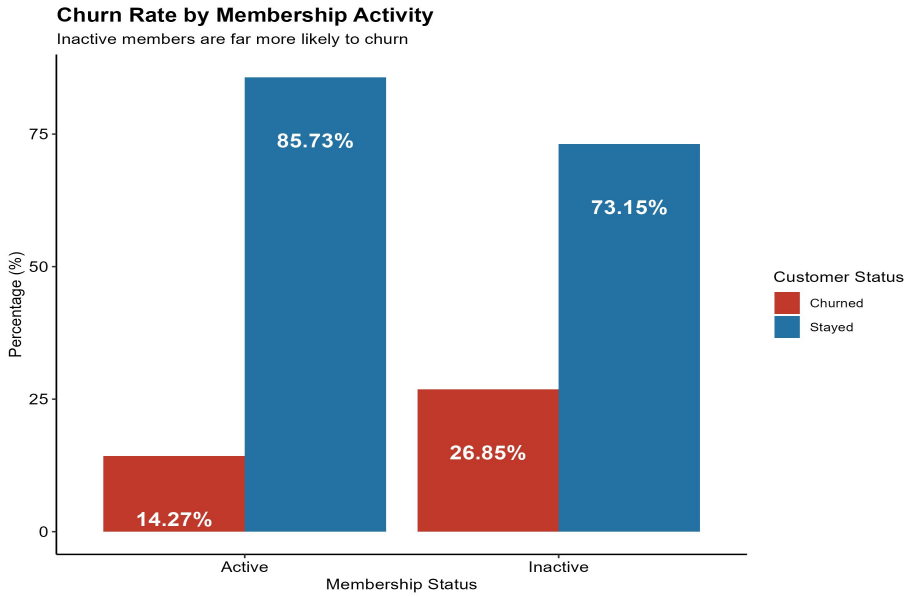

Figure 5. Churn rate comparison between active and inactive members.

Inactive members — those who are not regularly using the bank's services — churn at 26.85%, nearly double the 14.27% churn rate of active members. This is perhaps the most actionable finding in the entire analysis. Unlike age or country, activity level is something a bank can directly influence.

> *Re-engagement campaigns targeted at inactive members — such as personalized emails, fee waivers, or exclusive offers — could meaningfully reduce churn rates. If even half of the currently inactive churned customers could be converted to active status, the churn rate would drop significantly.*

Finding 6 — The 45 to 59 Age Group is the Highest Risk

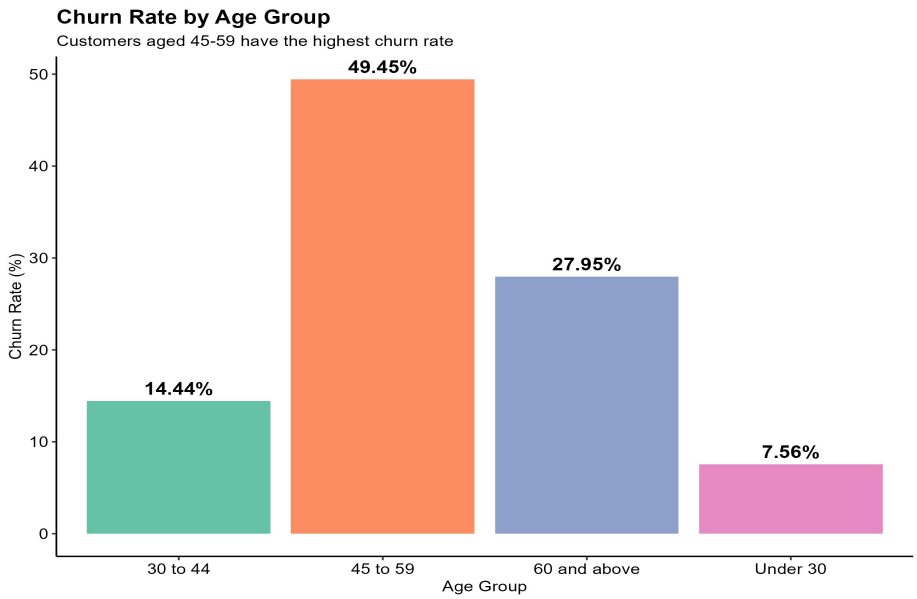

Figure 6. Churn rate broken down by age group.

When we group customers by age bracket, the 45 to 59 group has an alarming churn rate of 49.45% — meaning nearly half of all customers in this age group have left. The under-30 group is the most loyal at just 7.56%. This clear age gradient suggests the bank's products and services are well-suited to younger customers but fail to meet the evolving needs of middle-aged and older clients.

Finding 7 — Germany's Problem Runs Deep

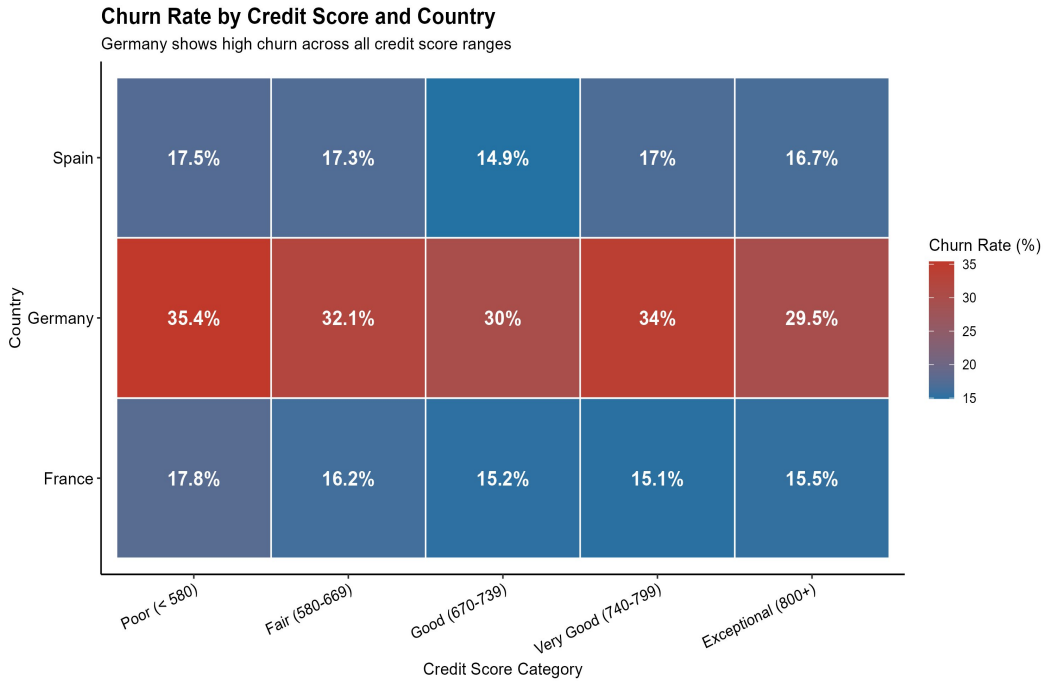

Figure 7. Heatmap of churn rate by country and credit score category.

This heatmap combines two variables — country and credit score — to reveal something remarkable. Germany's churn problem is not explained by poor credit scores. Even German customers with exceptional credit scores (800+) churn at 29.5%, while French customers with poor credit scores only churn at 17.8%. The entire Germany row is a deep red regardless of credit quality.

> *This rules out credit risk as an explanation for German churn. The cause must be structural — likely related to market competition, pricing strategy, or customer experience in the German market specifically.*

Finding 8 — A Shocking Product Paradox

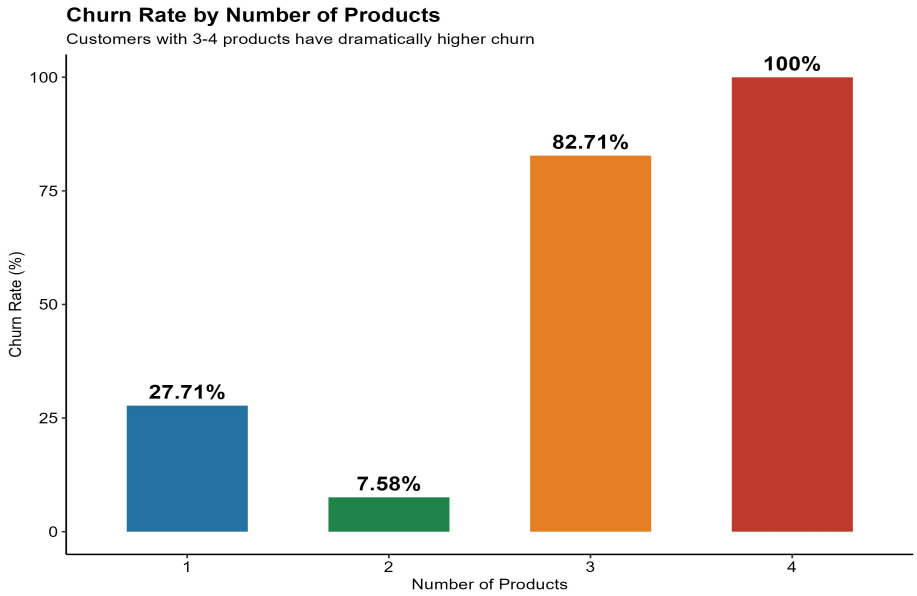

Figure 8. Churn rate by number of bank products held.

The most counterintuitive finding in this entire analysis: customers with 3 or 4 bank products churn at 82.71% and 100% respectively. You might expect that customers who hold more products are more committed — but the data says the opposite. Customers holding 2 products have the lowest churn rate at just 7.58%.

> *A possible explanation: customers forced into multiple products through bundling strategies may feel trapped rather than valued, leading to resentment and eventual departure. The 2-product sweet spot suggests this is the optimal level of engagement without over-selling.*

The High-Risk Customer Profile

Combining all findings, we can now describe exactly what a high-risk customer looks like compared to a customer who is likely to stay.

Table 2. Churned vs. Stayed Customer Comparison

Metric	Churned Customer	Stayed Customer	Key Difference
Average Age	44.8 years	37.4 years	7 years older
Average Credit Score	645	652	Slightly lower
Average Balance (USD)	$91,109	$72,745	$18,364 more
Average Tenure	4.9 years	5.0 years	Similar
Average Salary (USD)	$101,466	$99,738	Similar
% Female	55.9%	42.7%	More female
% Germany	40.0%	21.3%	2x more German
% Inactive	63.9%	44.5%	More inactive

The profile of a customer most likely to churn is: a woman in her mid-40s, based in Germany, with a healthy account balance, who has become inactive in her banking usage. She is not a problem customer — she is a valuable one who is quietly slipping away.

Conclusion and Recommendations

This analysis has uncovered five clear and actionable insights for bank management.

First, the 20.4% overall churn rate demands urgent attention. This is not a minor data anomaly — it represents a significant and ongoing loss of customers and revenue.

Second, Germany requires a dedicated investigation and intervention strategy. Churn rates above 30% across all credit score categories suggest a market-level problem that cannot be solved by individual-level retention tactics alone.

Third, the bank should launch targeted re-engagement campaigns for inactive members. This is the highest-leverage action available because activity level is directly controllable — unlike age or geography.

Fourth, customers aged 45 to 59 need a dedicated retention product. Whether this is a premium account tier, wealth management advisory, or retirement planning services, this age group needs

a compelling reason to stay.

Fifth, the bank should review its product bundling strategy. The dramatic increase in churn among 3 and 4 product customers suggests these customers feel over-sold rather than well-served.

Future Research Questions

#	Future Question
1	Does credit score truly not matter for German customers — or is there a threshold effect?
2	Are inactive members inactive by choice, or because the bank's digital experience is poor?
3	What specific products do 3 and 4 product customers hold that correlate with churn?
4	How does churn vary by tenure — do long-term customers eventually become more loyal?
5	Can a predictive churn model flag at-risk customers 3-6 months before they leave?

Works Cited

Kaggle. (2023). *Bank customer churn prediction dataset.* https://www.kaggle.com/datasets/shubhammeshram579/bank-customer-churn-prediction

R Core Team. (2024). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing. https://www.r-project.org/

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Firke, S. (2023). *janitor: Simple tools for examining and cleaning dirty data.* https://CRAN.R-project.org/package=janitor

Hadley, W. (2016). *ggplot2: Elegant graphics for data analysis.* Springer-Verlag. https://ggplot2.tidyverse.org